

Calculus I Lesson 18

Product Rule for Derivative: $D_x(f(x) \cdot g(x)) = f(x) \cdot g'(x) + g(x) \cdot f'(x)$

1) $f(x) = (x^3 + 4)(x - 7)$

a) $D_x(x^3 + 4) = \underline{\hspace{2cm}}?$

b) $D_x(x - 7) = \underline{\hspace{2cm}}?$

c) $f'(x) = \underline{\hspace{2cm}}?$

2) $f(x) = x^2 \cdot \sin x$

a) $D_x(x^2) = \underline{\hspace{1cm}}?$

b) $D_x(\sin x) = \underline{\hspace{1cm}}?$

c) $f'(x) = \underline{\hspace{1cm}}?$

$$3) f(x) = (x^2 + 5x)(2x^2 + 5x - 5)$$

$$\text{a) } D_x(x^2 + 5x) = \underline{\hspace{1cm}}?$$

$$\text{b) } D_x(2x^2 + 5x - 5) = \underline{\hspace{1cm}}?$$

$$\text{c) } f'(x) = \underline{\hspace{1cm}}?$$

4) $f(x) = (\cos x)(5 + \sin x)$

a) $D_x(\cos x) = \underline{\hspace{1cm}}?$

b) $D_x(5 + \sin x) = \underline{\hspace{1cm}}?$

c) $f'(x) = \underline{\hspace{1cm}}?$

5) $f(x) = x^2 \cdot \sin x$

a) $D_x(x^2) = \underline{\hspace{1cm}}?$

b) $D_x(\sin x) = \underline{\hspace{1cm}}?$

c) $f'(x) = \underline{\hspace{1cm}}?$

6) $f(x) = (x^{-2})(2x^2 + \sin x)$

a) $D_x(x^{-2}) = \underline{\hspace{2cm}}?$

b) $D_x(2x^2 + \sin x) = \underline{\hspace{2cm}}?$

c) $f'(x) = \underline{\hspace{2cm}}?$

7) $f(x) = (x^{2/3} + 1)(5x^2 + \sin x)$

a) $D_x(x^{2/3} + 1) = \underline{\hspace{1cm}}?$

b) $D_x(5x^2 + \sin x) = \underline{\hspace{1cm}}?$

c) $f'(x) = \underline{\hspace{1cm}}?$

8) $f(x) = (x^{-3/4} + 7x)(x^3 + \cos x)$

a) $D_x(x^{-3/4} + 7x) = \underline{\hspace{1cm}}?$

b) $D_x(x^3 + \cos x) = \underline{\hspace{1cm}}?$

c) $f'(x) = \underline{\hspace{1cm}}?$

9) $f(x) = (x^{-5/4} + 2x^3)(x^{1/2} + \cos x)$

a) $D_x(x^{-5/4} + 2x^3) = \underline{\hspace{2cm}}?$

b) $D_x(x^{1/2} + \cos x) = \underline{\hspace{2cm}}?$

c) $f'(x) = \underline{\hspace{2cm}}?$

